

# Looking to reduce board-space for 3-phase BLDC or PMSM designs?

High-integration gate driver IC with on-chip power module



3-Phase BLDC Motor Gate Driver  
With Power Module

The combination of the MCP8024 gate-driver with its integrated power module, and one of Microchip's PIC® MCUs or dsPIC® Digital Signal Controllers (DSCs), offers a complete and easy solution for driving three-phase BLDC or PMS motors in harsh environments.

The MCP8024 gate driver reduces board space for three-phase Brushless DC (BLDC) or Permanent Magnet Synchronous Motor (PMSM) designs by integrating all of the power, sensing and protection needed for a robust and highly efficient design.

In addition to the power module, voltage regulators, current-sensing amplifiers and over-current protection, the MCP8024 also offers an adjustable step-down DC-to-DC converter. This provides the flexibility to drive a broad range of MCUs, DSCs and FPGAs and to reduce time-to-market for new designs.

The MCP8024 is also the first motor driver IC to offer configurable safety parameters for driver dead-time management, driver blanking-time control, and Over-Current Limit (OCL) for external MOSFETs.

With thermally enhanced 40-pin QFN and 48-pin TQFP packages, and a wide operating temperature range from -40°C to +150°C, the MCP8024 can be used in harsh industrial and automotive applications.



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